

CISC 473: Deep Learning — Project Proposal

P22: Cross-Architecture Knowledge Distillation

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GitHub Repository: <https://github.com/l33zard/cakd.git>

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Problem Statement and Motivation

Knowledge distillation (KD) is the process of compressing a large, high-capacity model—the *teacher*—into a smaller, cheaper *student* model that preserves much of the teacher’s predictive accuracy. Rather than training the student on hard one-hot labels alone, KD trains it to match the teacher’s softened output distribution, which encodes the teacher’s learned similarity structure over classes [1]. Subsequent work showed that additional supervision can be extracted from intermediate representations rather than logits alone [2, 5].

The efficiency goal is concrete. Serving large models means routing billions of queries through energy-hungry data centres, and there is a strong push to move inference onto personal and embedded hardware—smartphones, laptops, wearables and automotive chips—so that features work instantly, offline, and without transmitting sensitive data. Such devices impose strict memory, latency and power budgets, and typically execute on CPUs or small accelerators where parameter count and FLOPs directly determine feasibility. Distillation allows a compact student to inherit teacher behaviour while cutting inference cost, making real-time deployment in robotics, autonomous systems and edge vision practical.

Cross-architecture KD is the harder setting in which teacher and student belong to different architectural families. We focus on the gap between convolutional neural networks (CNNs) and vision transformers (ViTs). CNNs encode images as hierarchical spatial feature maps produced by local convolutions and carry a strong locality and translation-equivariance prior; ViTs [11] encode images as sequences of patch tokens related by global self-attention. Their intermediate representations therefore differ in structure, dimensionality and inductive bias, so feature-level transfer methods designed for CNN-to-CNN pairs do not apply directly, and logit-only KD discards most of the teacher’s internal knowledge.

This project benchmarks established KD methods on CIFAR-100 and proposes an intermediate-feature alignment strategy for CNN-to-ViT transfer, evaluated under a matched parameter and FLOP budget and diagnosed using representational similarity analysis.

Literature Survey

Related Works

[1] Hinton, Vinyals & Dean, 2015. “Distilling the Knowledge in a Neural Network.” **NeurIPS Deep Learning Workshop**. **arXiv:1503.02531**. The paper introduces vanilla KD: the student minimises a weighted combination of the cross-entropy loss on ground-truth labels and a Kullback–Leibler divergence against the teacher’s logits softened by a temperature τ . The softened distribution exposes “dark knowledge”—the relative probabilities the teacher assigns to incorrect classes—and the authors demonstrate consistent gains for compact students on MNIST and large-scale speech recognition, as well as for distilling ensembles into single models. Vanilla KD is our first baseline and the logit-level loss term that our proposed alignment loss is added on top of; it is also the only listed method that is trivially architecture-agnostic, since it operates purely on output distributions.

[2] Tian, Krishnan & Isola, 2020. “Contrastive Representation Distillation.” **ICLR**. **arXiv:1910.10699**. CRD reformulates distillation as maximising a lower bound on the mutual information between teacher and student representations, using a contrastive objective that pulls together teacher–student features of the same input and pushes apart features of different inputs drawn from a memory bank. It outperforms vanilla KD and a broad suite of feature-matching methods on CIFAR-100 and ImageNet, including several teacher–student pairs with differing CNN families. CRD is a strong feature-level baseline for us; unlike our work, it compares pooled global embeddings and does not address the token-versus-spatial mismatch that arises when the student is a transformer.

[3] Zhao, Cui, Song, Qiu & Liang, 2022. “Decoupled Knowledge Distillation.” **CVPR**. **arXiv:2203.08679**. DKD decomposes the classical KD loss into target-class knowledge distillation (TCKD) and non-target-class knowledge distillation (NCKD), showing analytically that the standard formulation couples the NCKD term to the teacher’s confidence on the target class and thereby suppresses exactly the signal that carries most of the dark knowledge. Decoupling and independently reweighting the two terms recovers or exceeds state-of-the-art feature-based methods on CIFAR-100, ImageNet and MS-COCO at essentially no extra training cost. DKD serves as our strongest logit-only baseline and establishes the reference our cross-architecture method must beat to justify its additional alignment module.

[4] Chen, Liu, Zhao & Jia, 2021. “Distilling Knowledge via Knowledge Review.” **CVPR**. **arXiv:2104.09044**. ReviewKD argues that conventional feature distillation pairs layers at the same depth, whereas a student’s deeper layers can profitably “review” the teacher’s shallower features. It introduces cross-stage connections together with an attention-based fusion module and a hierarchical context loss to aggregate multi-level teacher features, yielding consistent improvements on classification, detection and segmentation. ReviewKD directly motivates our study of *which* layer pairs to align: because CNN and ViT depth semantics do not correspond, layer-pair selection becomes an explicit ablation axis in our work rather than a fixed design choice.

[5] Romero, Ballas, Kahou, Chassang, Gatta & Bengio, 2015. “FitNets: Hints for Thin Deep Nets.” **ICLR**. **arXiv:1412.6550**. FitNets is the original intermediate-feature distillation method: a “hint” layer in the teacher supervises a “guided” layer in the student through an L_2 loss, with a learnable regressor inserted to reconcile the mismatched feature dimensionalities before the standard KD objective is applied. This made it possible to train students that are deeper and

thinner than their teachers. Our projection module is a direct descendant of the FitNets regressor, generalised from a channel-dimension mismatch between two CNNs to a structural mismatch between a spatial feature map and a patch-token sequence.

[6] Kornblith, Norouzi, Lee & Hinton, 2019. “Similarity of Neural Network Representations Revisited.” ICML. arXiv:1905.00414. The paper analyses measures of representational similarity and shows that Centered Kernel Alignment (CKA) is invariant to orthogonal transformation and isotropic scaling while remaining sensitive to genuine differences, unlike CCA variants; this makes it reliable for comparing layers of different widths or of independently trained networks. We adopt CKA both as a diagnostic—quantifying how CNN-teacher and ViT-student representations relate before and after distillation—and, optionally, as a differentiable alignment objective.

[7] Park, Kim, Lu & Cho, 2019. “Relational Knowledge Distillation.” CVPR. arXiv:1904.05068. RKD transfers the *structure* of the teacher’s embedding space rather than individual activations, penalising discrepancies in pairwise distances and triplet angles among samples in a mini-batch. Because it constrains only relations between examples, it is insensitive to the absolute geometry and dimensionality of either representation space. This property is attractive for cross-architecture transfer and gives us a natural secondary comparison for our projection-based alignment.

Open Problem

Existing feature-level distillation methods [2, 4, 5, 7] are developed and evaluated almost exclusively on CNN-to-CNN pairs, while the methods that transfer cleanly across architectures [1, 3] use logits only and therefore discard intermediate knowledge. Cross-architecture KD with a CNN teacher and a ViT student consequently lacks (i) a systematic study of how to align spatial feature maps with patch-token sequences, including which layer pairs should be matched when depth semantics do not correspond; (ii) an efficiency analysis that reports accuracy jointly with parameters, FLOPs and training cost under a matched budget, as required for CPU and edge deployment; and (iii) a principled diagnosis of *why* transfer succeeds or fails—temperature and loss-weighting ablations are rarely reported for this setting, and CKA-based analysis of teacher–student representational similarity [6] has not been applied to it. This project addresses all three gaps.

Proposed Methodology

Baseline

Dataset. All experiments use CIFAR-100 [12]: 60,000 32×32 colour images across 100 classes, with the standard 50,000/10,000 train/test split. We hold out 5,000 training images as a validation set for hyperparameter and checkpoint selection. Standard augmentation (random crop with 4-pixel padding, horizontal flip, per-channel normalisation) is applied. If compute permits, we will scale the best configuration to TinyImageNet (64×64 , 200 classes) to test whether conclusions hold at higher resolution.

Models. The CNN-to-CNN teacher is ResNet-110 [8] trained on CIFAR-100 with the standard recipe; the corresponding student is MobileNetV2 [9], giving a conventional compression baseline. For the cross-architecture setting we use a ResNet-50 [8] teacher, whose wider stage outputs are better matched to a token embedding dimension, and a DeiT-Tiny [10] student (192-dimensional embedding, 12 layers, 3 heads) adapted to 32×32 inputs with a reduced patch size.

Reproduced baselines. We will reproduce Vanilla KD [1], CRD [2], DKD [3] and ReviewKD [4] starting from the RepDistiller framework and the authors’ official releases, retaining their published hyperparameters where available so that reported numbers can be validated before we modify anything.

Metrics. Top-1 test accuracy is the primary metric; we additionally report top-5 accuracy, parameter count, inference FLOPs and wall-clock training time per run, so that any accuracy gain can be read against its cost.

Proposed Extension

CNN feature maps $F_T \in \mathbb{R}^{C \times H \times W}$ and ViT token sequences $F_S \in \mathbb{R}^{N \times D}$ cannot be compared directly. We propose a lightweight intermediate-feature alignment module that operates entirely outside the student, leaving the DeiT architecture unmodified. The teacher map is spatially pooled or bilinearly interpolated to a $\sqrt{N} \times \sqrt{N}$ grid, flattened into N vectors, and passed through a learnable 1×1 projection $\mathbb{R}^C \rightarrow \mathbb{R}^D$. The resulting pseudo-tokens are aligned with the student’s tokens by an alignment loss $\mathcal{L}_{\text{align}}$, instantiated either as a differentiable CKA objective [6] or as an attention-map transfer loss that matches the teacher’s spatial activation statistics to the student’s self-attention maps. The total objective is

$$\mathcal{L} = (1 - \alpha) \mathcal{L}_{\text{CE}} + \alpha \tau^2 \mathcal{L}_{\text{KD}}(\tau) + \beta \mathcal{L}_{\text{align}}.$$

Because the projection is discarded at inference, the deployed student’s parameter count and FLOPs are unchanged.

Hypothesis. Explicit CKA or attention-based alignment of intermediate representations gives the DeiT-Tiny student supervision that logit-only KD cannot provide, improving top-1 accuracy over Vanilla KD, CRD, DKD and ReviewKD at a matched parameter and FLOP budget.

Ablations. We will sweep the temperature $\tau \in \{1, 2, 4, 8, 16\}$, the distillation weight α , the alignment weight β , the choice of alignment loss, and the teacher–student layer pairs (early, middle, late and multi-layer combinations), following the layer-pairing question raised by ReviewKD [4].

Evaluation Plan

- **Dataset and splits.** CIFAR-100 with the standard train/test split and a fixed 5,000-image validation subset held out from training; identical splits, augmentation and schedules across all methods. Optional TinyImageNet scale-up.
- **Baselines.** DeiT-Tiny and MobileNetV2 trained from scratch (lower bound), Vanilla KD [1], CRD [2], DKD [3], ReviewKD [4], and the teacher’s own accuracy (upper bound).
- **Metrics.** Top-1 and top-5 accuracy, parameters, FLOPs, training time, and layer-wise CKA similarity between teacher and student measured before and after distillation.
- **Statistical reliability.** Every main result uses at least three independent runs with fixed seeds, reported as mean $\pm$ standard deviation; ablation sweeps use a single seed for screening, with the selected configurations rerun over three seeds.

Early Novel Contribution (Optional Bonus)

We propose two concrete contributions. First, a systematic CKA-based study of CNN-to-ViT knowledge distillation: layer-wise CKA heatmaps between a ResNet teacher and a DeiT-Tiny student, measured before and after distillation under each baseline method, which to our knowledge

has not been reported and would give a diagnostic account of where cross-architecture transfer breaks down. Second, a lightweight intermediate-feature alignment module that requires no modification to the ViT architecture and adds no inference-time cost, since the projection is used only during training. If the ablations show that alignment improves student accuracy under a matched parameter and FLOP budget, and that the improvement is accompanied by measurably higher CKA similarity, we will submit this as our +2 bonus contribution.

Project Timeline and Roles

Week	Tasks	Responsible
1–2	Read DKD, CRD and ReviewKD; environment and repository setup; CIFAR-100 data pipeline; train ResNet-110 teacher and from-scratch student baselines; Proposal due	Lee Xuan & Fong Yee Xin
3	Vanilla KD implementation; temperature τ and distillation-weight α ablations	Lee Xuan
4	CRD distillation baseline; memory-bank and negative-sampling configuration	Fong Yee Xin
5	DKD distillation baseline; ResNet-50 and DeiT-Tiny setup for cross-architecture ResNet-50 $\rightarrow$ DeiT-Tiny transfer	Fong Yee Xin
6	ReviewKD baseline; layer-wise CKA analysis before and after distillation; Midterm due	Lee Xuan
7	Intermediate-feature alignment module for the CNN $\rightarrow$ ViT case; projection and alignment-loss implementation	Fong Yee Xin
8	Extended experiments and ablations (β , layer pairs, alignment loss); efficiency analysis of accuracy vs. FLOPs	Lee Xuan & Fong Yee Xin
9	Results analysis; accuracy tables with mean $\pm$ std; CKA heatmaps; report writing	Lee Xuan
10	Final report, reproducibility checks, README and code release; Final due	Lee Xuan & Fong Yee Xin

Contribution Declaration

Member	Primary Responsibilities	Estimated %
Lee Xuan (Project Lead)	Project coordination, literature review, Vanilla KD and ReviewKD baselines, CKA analysis, efficiency analysis, report writing	50%
Fong Yee Xin (Engineering Lead)	Data and training pipeline, CRD and DKD baselines, cross-architecture setup, feature-alignment module implementation, ablation design and execution, evaluation code and README	50%

References

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